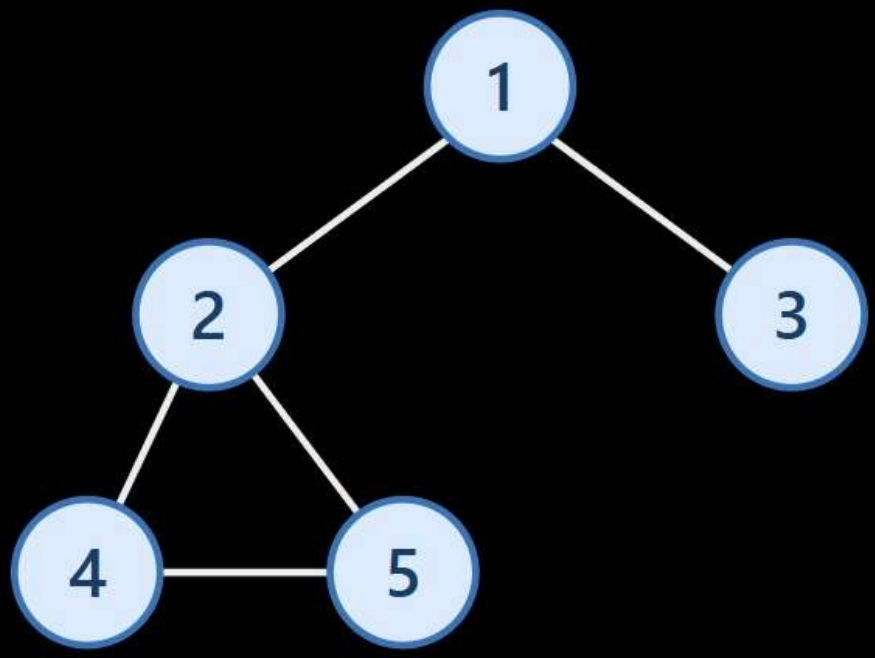
Question 1: BFS and DFS

Diagram



Programiz Python Online Compiler

main.py

Visualize Run ▶

```python

from collections import deque

graph = {
    1: [2, 3],
    2: [1, 4, 5],
    3: [1],
    4: [2, 5],
    5: [2, 4]
}

def bfs(start):
    visited = {start}
    queue = deque([start])
    while queue:
        node = queue.popleft()
        print(node, end=" ")
        for neighbour in graph[node]:
            if neighbour not in visited:
                visited.add(neighbour)
                queue.append(neighbour)

def dfs(node, visited):
    visited.add(node)
    print(node, end=" ")
    for neighbour in graph[node]:
        if neighbour not in visited:
            dfs(neighbour, visited)
```

Output

```
BFS:
1 2 3 4 5
DFS:
1 2 4 5 3
```

```
            dfs(neighbour, visited)

print("BFS:")
bfs(1)

print("\nDFS:")
dfs(1, set())
print()
```